

Does Dealer Gamma Move Volatility? Causal Evidence from the Option-Expiration Roll-off

Code Documentation: scripts, data flow, and pipeline

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Abstract

This document describes the analysis code for an *empirical* instrumental-variables study of how option-dealer gamma hedging affects equity-market volatility — whether dealer gamma causally moves realized volatility, estimated on WRDS / OptionMetrics data. The dollar gamma scheduled to expire at each monthly option expiration serves as a calendar-determined instrument for dealer gamma. Each script is documented below by purpose, inputs, and outputs, and the overall pipeline is shown as a diagram. The licensed source data are not included in the repository; the acquisition scripts rebuild every panel from WRDS for users with their own access.

Code availability. github.com/duc-v-le/dealer-gamma-volatility

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1 Overview

The repository is organized as a single pipeline: WRDS / OptionMetrics panels are acquired, processed, and analyzed, producing the result tables in **data/** and figures in **figures/**.

Empirical workflow (real data)

- Data Acquisition:
 - Scripts named `wrds_download_*.py` connect via `wrds_lib.py`.
 - They pull *dealer-gamma* and *price panels* from WRDS / OptionMetrics.
 - Raw files are saved directly to the **data/real/** directory.
- Data Analysis:
 - Scripts named `e4_*.py` read the downloaded panels.
 - They estimate *instrumental-variables* and *panel specifications*.
 - Final outputs are written as small *result tables* and *figures*.

Data and credentials

- Authentication:
 - The WRDS username and password are read at run time from `~/.pgpass` or the `WRDS_USER` environment variable.
 - No secret keys or credentials are stored directly in the code source files.
- Data Privacy:
 - The licensed data panels saved in **data/real/** are strictly non-redistributable.
 - Only the aggregated final result tables are preserved inside **data/**.

2 Pipeline diagram

Figure 1 shows the data flow. Boxes are colour-coded:

- **External source** (red): licensed WRDS / OptionMetrics panels.
- **Processing scripts** (blue): acquisition and analysis scripts.
- **Intermediate data** (gray): panels and other temporary files.
- **Final outputs** (green): result tables and figures.

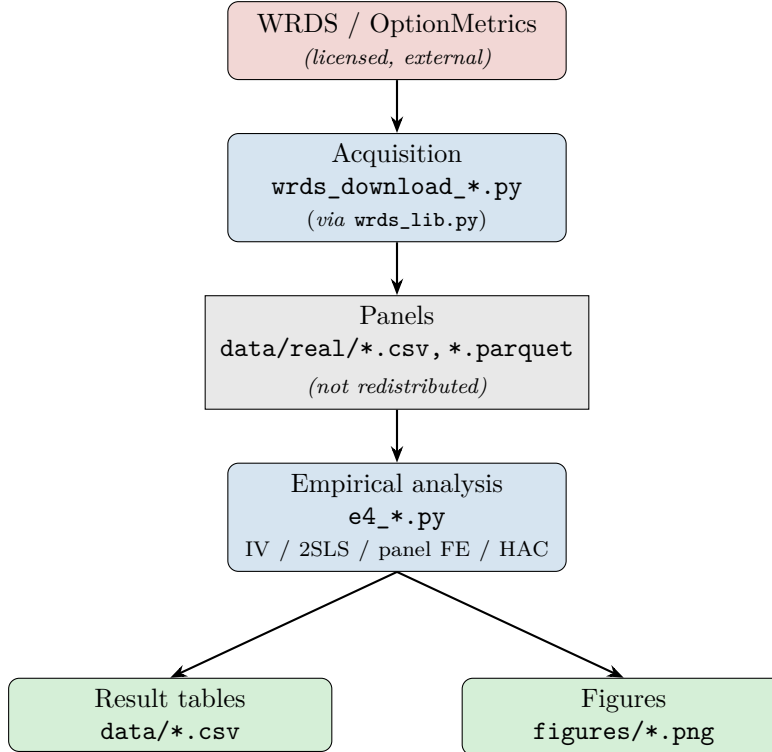


Figure 1: End-to-end pipeline. WRDS / OptionMetrics data is acquired, processed, and analyzed top-to-bottom, producing the result tables and figures.

Module dependencies.

- `wrds_lib.py` is imported by every acquisition and WRDS-backed analysis script.
- `paperA_figures.py` reads several result tables to assemble the manuscript figures.
- All other scripts are standalone entry points.

3 Data-acquisition layer

These scripts connect to WRDS and write panels to `data/real/`. They require a WRDS account; the panels they produce are the inputs to the empirical layer.

Script	What it does	Writes
<code>wrds_lib.py</code>	Minimal WRDS PostgreSQL connection helper; reads credentials from <code>~/.pgpass</code> . Imported by every WRDS script.	— (library)
<code>wrds_schema_probe.py</code>	Documents OptionMetrics schema discovery (tables, columns) for provenance.	— (console)
<code>wrds_rank_universe.py</code>	Ranks the optionable universe by liquidity to define the sample.	<code>universe_ranked.csv</code>
<code>wrds_download_spx_gamma.py</code>	Daily SPX dealer-gamma and price panel (OptionMetrics SPX).	<code>spx_gamma_panel.csv</code>
<code>wrds_download_gamma.py</code>	Generalized single-name gamma panel for any ticker.	<code><name>_gamma_panel.csv</code>
<code>wrds_download_panel.py</code>	Top-50 liquid-options panel of net dealer gamma.	<code>panel_gamma.csv</code>
<code>wrds_download_panel_sp500.py</code>	Full S&P 500 panel; parallel-by-year, resumable.	<code>panel_sp500.parquet</code> , <code>sp500_secids.csv</code>
<code>wrds_download_panel_sp500_cp.py</code>	Full S&P 500 panel with call/put-split gamma.	<code>panel_sp500_cp.parquet</code>
<code>wrds_download_spx_expiry.py</code>	SPX gamma split by days-to-expiry bucket.	<code>spx_gamma_expiry.csv</code>
<code>wrds_download_spx_rolloff.py</code>	SPX dollar-gamma scheduled to expire by the next monthly expiration (the instrument).	<code>spx_rolloff.csv</code>
<code>wrds_download_rolloff.py</code>	Generalized scheduled-roll-off panel for any ticker.	<code><name>_rolloff.csv</code> , <code>spx_rolloff.csv</code>

4 Empirical-analysis layer

These scripts read the panels above and estimate the specifications. Each writes a small result table (regression coefficients, t -statistics, first-stage diagnostics); some also write a figure. The common dependent variable is forward realized volatility; the key regressor is dealer gamma, instrumented by the gamma scheduled to roll off at the next monthly expiration.

Script	What it does	Reads → writes
e4_leadlag_real.py	Lead-lag of dealer short-gamma concentration vs. future realized volatility on the real SPX panel.	spx_gamma_panel.csv → e4_real_leadlag.csv, e4_real_leadlag.png
e4_real_robust.py	Newey–West (HAC) inference plus a stress-episode overlay.	spx_gamma_panel.csv → e4_real_hac.csv, e4_real_episodes.png
e4_crossname.py	Single-name generalization (e.g. NVDA/TSLA/AAPL) with a VIX-level control.	<name>_gamma_panel.csv → e4_crossname.csv, e4_crossname.png
e4_causal_opex.py	Expiration-calendar (OPEX) instrument for dealer gamma; hand-rolled 2SLS plus an event study.	spx_gamma_panel.csv → e4_causal_opex.csv, e4_opex_eventstudy.png
e4_iv2sls.py	Proper 2SLS (linearmodels) on gamma-by-expiry; diagnoses instrument strength.	spx_gamma_expiry.csv → e4_iv2sls.csv
e4_iv_rolloff.py	Scheduled-roll-off IV (the strong instrument).	spx_rolloff.csv → e4_iv_rolloff.csv
e4_iv_rolloff_fwdvol.py	Roll-off IV under the unified forward-volatility outcome.	spx_rolloff.csv → e4_iv_rolloff_fwdvol.csv
e4_iv_robust_excl.py	Exclusion-restriction hardening: volume and Δ VIX controls, over-identification J , falsification placebos.	spx_rolloff.csv → e4_iv_robust_excl.csv
e4_paperA_outcomes.py	Roll-off IV across single names and outcomes (volatility, tail, variance, kurtosis).	<name>_rolloff.csv → e4_paperA_outcomes.csv
e4_panel_iv.py	Pooled panel 2SLS with time fixed effects and liquidity heterogeneity (top-50).	panel_gamma.csv → e4_panel_iv.csv
e4_panel_mechanism.py	Convexity-near-stress test across VIX regimes (full S&P 500).	panel_sp500.parquet → e4_panel_mechanism.csv
e4_paperA_robustness.py	Specification robustness: peak-0DTE subsample and OPEX-week exclusion.	panel_sp500.parquet → e4_paperA_robustness.csv

Script	What it does	Reads → writes
e4_horizon.py	Forward-volatility horizon robustness (1/5/10/21-day).	panel_sp500.parquet → e4_horizon.csv
e4_systemic_aggregate.py	Aggregate market-gamma → index-volatility IV and a systemic-regime decomposition.	panel_sp500.parquet, spx_gamma_panel.csv → e4_systemic_aggregate.csv
e4_sign_robust.py	Call/put gamma decomposition; validates the dealer-sign convention.	panel_sp500_cp.parquet → e4_sign_robust.csv
e4_twoway_fullpanel.py	Two-way (firm + date) clustered panel outcomes.	panel_sp500_cp.parquet → e4_twoway_fullpanel.csv
e4_earnings_exclusion.py	Re-runs the panel IV excluding earnings windows (exclusion-restriction defense).	panel_sp500_cp.parquet → e4_earnings_exclusion.csv

5 Utilities and manuscript figures

Script	What it does	Reads → writes
<code>paperA_figures.py</code>	Assembles the eight manuscript figures (F1–F8) from the result tables and panels.	result tables, <code>panel_sp500_cp.parquet</code> → <code>paperA_F1..F8.png</code>

6 Output artifacts

data/ (in the repository).

Aggregated result tables — small CSVs of coefficients, t -statistics, and summary statistics.
No licensed records.

figures/ (in the repository).

Generated PNG figures: the real-data plots (`e4_real_*`, `e4_opex_eventstudy`, `e4_crossname`) and the manuscript figures (`paperA_F1..F8`).

data/real/ (not in the repository).

The licensed WRDS / OptionMetrics panels. Rebuilt from source by the acquisition scripts for users with their own WRDS access.

7 Reproduction order

All analysis requires a WRDS account. Run the acquisition script that produces a panel before the analysis script that consumes it. The main dependencies are:

Acquisition script	Analysis scripts (run after)
<code>wrds_download_spx_gamma.py</code>	<code>e4_leadlag_real.py</code> , <code>e4_real_robust.py</code> , <code>e4_causal_opex.py</code>
<code>wrds_download_spx_rolloff.py</code>	<code>e4_iv_rolloff.py</code> , <code>e4_iv_rolloff_fwdvol.py</code> , <code>e4_iv_robust_excl.py</code>
<code>wrds_download_spx_expiry.py</code>	<code>e4_iv2sls.py</code>
<code>wrds_download_gamma.py</code>	<code>e4_crossname.py</code>
<code>wrds_download_rolloff.py</code>	<code>e4_paperA_outcomes.py</code>
<code>wrds_download_panel.py</code>	<code>e4_panel_iv.py</code>
<code>wrds_download_panel_sp500.py</code>	<code>e4_panel_mechanism.py</code> , <code>e4_paperA_robustness.py</code> , <code>e4_horizon.py</code> , <code>e4_systemic_aggregate.py</code>
<code>wrds_download_panel_sp500_cp.py</code>	<code>e4_sign_robust.py</code> , <code>e4_twoway_fullpanel.py</code> , <code>e4_earnings_exclusion.py</code>

The sample-selection helper `wrds_rank_universe.py` and the schema probe `wrds_schema_probe.py` are run once to document universe construction and data provenance.